

Abdelrahman Essam

Software Engineering Student | Full-Stack & Systems Developer

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Summary

Computer & Communication Engineering student (GPA 3.76/4.0, top 20 in cohort) applying competitive-programming problem-solving to scalable full-stack systems. Builds and **load-tests** production-grade apps with Java, Spring Boot, React/Next.js, NestJS, MySQL, Kafka, and Docker. Seeking a software engineering internship or junior developer role.

Education

Faculty of Engineering, Alexandria University

Oct 2022 – July 2027

B.Sc. in Computer and Communication Engineering

Alexandria, Egypt

– **GPA: 3.76/4.0** — Ranked in the top 20 of the cohort.

– **Relevant Coursework:** Algorithms, Data Structures, OS, Computer Networks, Databases, Software Engineering, OOP, Multithreading, Design Patterns.

Technical Skills

Languages: Java, C++, C, Python, JavaScript, Typescript, Bash, C# **Frontend:** React.js, Next.js, HTML5, CSS3, Tailwind CSS

Backend: Spring Boot, Spring Security, NestJS, .NET, RESTful APIs, JWT, JPA/Hibernate, Microservices Architecture

Databases: MySQL, SQL, Postgres, Relational Database Design, Query Optimization

DevOps & Tools: Docker, AWS (EC2, RDS), Nginx, Linux, Apache Kafka, Zookeeper, pgBoss, Git, GitHub Actions, CI/CD, Maven, JUnit, Mockito, JaCoCo, Jest, JMeter

Core CS: Data Structures, Algorithms, OOP, Design Patterns, Multithreading, POSIX, RBAC, WebSockets

Projects

RentSphere — Rental Platform | GitHub | *Spring Boot, React.js, MySQL, AWS, Docker, JMeter*

Oct 2025 – Jan 2026

- Built a full-stack rental marketplace: **27 RESTful APIs**, **3-role Spring Security RBAC (JWT, BCrypt)**, with search, rental requests, contracts, and **PayPal** installment checkout.
- Modeled **8 MySQL tables** and tuned indexing **18 → 12** by **EXPLAIN**-verifying every query path on a seeded **100k-listing** dataset.
- **JMeter** load tests (25–200 concurrent users) exposed **N+1** image queries on listing endpoints; batched them with chunked **IN** fetches, cutting mixed-workload mean latency from **1.7 s to 42 ms** at 50 users.
- System sustains **200 concurrent users** at **0%** errors (≈ 71 req/s); at 100 users, **p95 $\approx$ 0.3 s** overall, listing-detail **p95 $\approx$ 15 ms** against 100k rows.
- Deployed a **4-service Docker Compose** stack (**MySQL 8, Spring Boot, React, Nginx**) on **AWS EC2 + RDS** with **rate limiting**, gzip, security headers, and **Actuator** health/metrics.
- Automated CI/CD via **GitHub Actions**: **233 tests** (**182 JUnit/Mockito**, **51 Jest**), **JaCoCo** 72% line coverage, and **Docker** build gates.

IoT Weather Station Monitoring | GitHub | *Java, Apache Kafka, Zookeeper, Docker*

Feb 2026 – May 2026

- Engineered a concurrent acquisition system in **Java** ingesting 10 independent IoT stations via multi-threaded executor pools.
- Streamed real-time data into **Apache Kafka** (**acks=all**, idempotent) through a REST adapter on the **Open-Meteo API**; containerized with multi-stage **Docker** builds.

Social Media Application | GitHub | *Spring Boot, React.js, MySQL, JWT, WebSockets, Apache JMeter*

May 2025 – Sep 2025

- Produced **45+ RESTful APIs** with role-based, stateless **JWT** security and real-time **WebSocket** (STOMP) messaging for 100+ concurrent users.
- Eliminated bottlenecks via **JPA fetch optimization**; **JMeter** tests cut average response time from **80 ms to 35 ms**.

Experience

Mylo | finX Squad | *React, Next.js, NestJS, TypeScript, pgBoss*

Aug 2026 – Present

Full-Stack Software Engineering Intern

Egypt

- Shipped to **production** a consumer-search and loan rescheduling view in **React/Next.js**, letting internal agents reschedule multiple loans per consumer in one flow — handling time cut from **15 to 9 minutes**.
- Fixed a **NestJS** metadata bug that dropped stored loan updates from the **LMS** service, restoring correct processing for **100%** of affected financial requests.
- Re-architected the fix around **pgBoss** job queues and shipped it to **production**: **500+ jobs/day** processed with zero dropped jobs.
- Migrated **10 UI components** to the **MyloKit** design system and **3 BFF endpoints** to **Next.js Server Actions**, removing an intermediate API layer.

EJADA Systems | GitHub | *Java, SQL, Spring Boot*

Jul 2026 – Aug 2026

Software Engineering Intern

Egypt

- Designed relational schemas and **20+** complex SQL queries for enterprise-style data models; built Java modules exercising **multithreading** and **exception handling** under Agile code review.

Achievements

- **ECPC 2025 Finalist** — Ranked **98th nationally** (Egyptian Collegiate Programming Contest); **1,000+** problems on LeetCode & Codeforces (graphs, DP, system design).